

DATA STRUCTURES

LAB PROGRAMS

24.Kruskal's algorithm

```
#include <stdio.h>

struct Edge {
    int u,v,w;
};

int parent[10];

int find(int i) {
    while(parent[i])
        i=parent[i];
    return i;
}

int unionSet(int i,int j) {
    if(i!=j) {
        parent[j]=i;
        return 1;
    }
    return 0;
}

int main() {
    struct Edge edge[5] = {
        {0,1,10},
```

{0,2,6},

{0,3,5},

{1,3,15},

{2,3,4}

};

int n=4,count=0,cost=0;

printf("Edges in MST:\n");

while(count < n-1) {

int min=999,a=-1,b=-1,index=-1;

for(int i=0;i<5;i++) {

if(edge[i].w < min) {

min=edge[i].w;

a=edge[i].u;

b=edge[i].v;

index=i;

}

}

int u=find(a);

int v=find(b);

if(unionSet(u,v)) {

printf("%d - %d : %d\n",a,b,min);

cost += min;

count++;

}

```
    edge[index].w = 999;  
}  
printf("Total Cost = %d",cost);  
return 0;  
}
```

Output

Edges in MST:

2 - 3 : 4

0 - 3 : 5

0 - 2 : 6

Total Cost = 15

=== Code Execution Successful ===